Practical No: 2

1. Aim: To Write A Program Using IF ELSE ELIF STATEMENT To Know The Greatest Number.

➢ Input:

- a=int(input('enter first number:'))
- b=int(input('enter second number:'))
- c=int(input('enter third number:'))
- if a>b:
- print('first number is greater than b')
- elif a>c:
- print('second number is greatest among all them')
- else:
- print('c is the largest among all')

➢ Output:

```
>
===== RESTART: C:\Users\User\AppData
enter first number:50
enter second number:100
enter third number:150
c is the largest among all
>
```

2. Aim: To Write A Program To Grade Students As Per There Marks Using IF ELSE ELIF Input Function.

- Input:
  - val1=int(input('enter marks of the first subject:'))
  - val2=int(input('enter marks of the second subject:'))
  - val3=int(input('enter marks of the third subject:'))
  - val4=int(input('enter marks of the fourth subject:'))
  - val5=int(input('enter marks of the fifth subject:'))
  - avg=int(val1+val2+val3+val4+val5)//5
  - if(avg>=90):
  - print('Grade A')
  - elif(avg>=80 & avg<90):
  - print('Grade B')
  - elif(avg>=70 & avg<80):
  - print('Grade C')
  - elif(avg>+60 & avg<70):
  - print('Grade D')
  - else:
  - print('Grade F')

- Output:

```
Enter "help" below or click "Help" abov

===== RESTART: C:\Users\User\AppData\Lc
enter marks of the first subject:87
enter marks of the second subject:82
enter marks of the third subject:92
enter marks of the fourth subject:86
enter marks of the fifth subject:96
Grade B
```

3. Aim: To Verify Whether Any Number Is Odd Or Even By Using IF ELSE Loop.

- Input:
  - val1=int(input('enter any number:'))
  - if(val1%2==0):
  - print('the given number is even')
  - else:
  - print('the given number is odd')

- Output:

```
===== RESTART: C:\Users\User\AppData\
enter any number:81
the given number is odd
```

4. Aim: The Write A Program To Compare Various Ages By Using Nested If Statement.

- Input:
  - tonny_age=43
  - jack_age=32
  - rose_age=29
  - if tonny_age>jack_age:
  - print('tonny is older than jack')
  - if tonny_age>rose_age:
  - print('tonny is oldest')

- Output:

```
===== RESTART: C:\Users\User\AppData\Local\Program
tonny is older than jack
tonny is oldest
```

5. Aim: To Write A Program To Display The Table Of Any Given Number By Using Range Function.

- Input:
  - n=int(input('enter any number:'))
  - for i in range(1,11):
  - print(n,'x',i,'=',n*i)

- Output:

```
===== RESTART: C:\Users\User\AppData\Local\Prog
enter any number:16
16 x 1 = 16
16 x 2 = 32
16 x 3 = 48
16 x 4 = 64
16 x 5 = 80
16 x 6 = 96
16 x 7 = 112
16 x 8 = 128
16 x 9 = 144
16 x 10 = 160
```

6. Aim: To Write A Program To Find Factorial Of Any No By Using Input Function And Range.

- Input:
  - n=int(input('enter any no:'))
  - factorial=1
  - for i in range(1,n+1):
  - factorial*=i
  - print(factorial)

- Output:

```
===== RESTART: C:\Users\User\AppData\Local\Programs\Pyt
enter any no:5
1
2
6
24
120
```

7. Aim: To Perform Various Operation By Using While Else Loop.

a. To find square of the given number:

- Input:
  - number=1
  - while number<5:
  - print(number, “has its square as”, number**2)
  - number=number+1
  - else:
  - print("no number left")
- Output:

```
>>
===== RESTART: C:\Users\User\AppData\Local\
1 has its square as 1
2 has its square as 4
3 has its square as 9
4 has its square as 16
no number left
>>
```

b. to write a program to break the while else loop:

- Input:
  - number=1
  - while number<=5:
  - print(number, ‘has its square as’, number**2)
  - number=number+1
  - if number==4:
  - break
- Output:

```
>>
===== RESTART: C:\Users\User\AppI
1 has its square as 1
2 has its square as 4
3 has its square as 9
>>
```

c. To continue the while else loop even after skipping a valid condition for the while loop:

- Input:
  - number=0
  - while number<6:
  - number=number+1
  - if number==4:
  - continue
  - print(number,'has its square as',number**2)

- Output:

```
>>
===== RESTART: C:\Users\User\AppData\Local'
1 has its square as 1
2 has its square as 4
3 has its square as 9
5 has its square as 25
6 has its square as 36
>>
```

8. Aim: To Write A Program For Finding Factorial Using While Loop.

- Input:
  - s=1
  - n=int(input('enter any number:'))
  - while n>1:
  - s=s*n
  - n=n-1
  - print(s)
- Output:

```
>>
===== RESTART: C:\Users\User\AppData\Lo
enter any number:4
4
12
24
>>
```

9. Aim: To Write A Program To Know Reversal Of Any Number By Using While Loop.

- Input:
  - number=int(input('enter any number'))
  - reverse=0
  - while number>0:
  - remainder=number%10
  - reverse=(reverse*10)+remainder
  - number=number//10
  - print('reverse of the entered number is',reverse)
- Output:

```
>>
===== RESTART: C:\Users\User\AppData\Local\P
enter any number31
reverse of the entered number is 1
reverse of the entered number is 13
>>
```

Practical No: 3

1. Aim: To Write A Program To Print Fibonacci Series Using For Range Function.

- Input:
  - n1 = 0
  - n2 = 1
  - num = int(input("Enter a No: "))
  - while n2 < num:
  - print(n2)
  - n1,n2 = n2,n1 + n2
- Output:

```
>
==== RESTART: C:\Users\User\AppData\Loca
Enter a No: 15
1
1
2
3
5
8
13
>
```

2. Aim : To Write A Program To Print A Pyramid Of * Using Range Function

- Input:
  - num=int(input(" enter the numbers of rows: "))
  - for i in range(1,num+1):
  - for j in range(1,i+1):
  - print("*", end='')
  - print('\n')
- Output:

```
>
==== RESTART: C:\Users\User\AppData\Loca
 enter the numbers of rows : 4

 *

 **

 ***
>
 ****
```

3. Aim: To Write Various Program for Manipulating List by Adding, Updating, Deleting, and Retrieving In a List.

- Input:
  - Tanmay= ['bob','Marry']
  - Tanmay.append('Bob')
  - print(Tanmay)
  - list1=['Aditya','Sahil']
  - list1.insert(0,'Sachin')
  - print(list1)
  - list2=['ABC','Tanmay']
  - list2[0]='Sachin'
  - print(list2)
  - list3=['ABC,'Tanmay']
  - list3.remove('ABC')
  - print(list3)
  - print(list2[1])
- Output:

```
>
==== RESTART: C:\Users\User\AppData\Local\Progra
['bob', 'Marry', 'Bob']
['Sachin', 'Aditya', 'Sahil']
['Sachin', 'Tanmay']
['Tanmay']
Tanmay
>
```

4. Aim: To Write Various Program For Manipulating Tuples By Typecasting.

- Input:
  - Adtiya= ('bob','Marry')
  - Adtiya_list=list(Adtiya)
  - Adtiya_list[1]='Adtiya'
  - Adtiya=tuple(Adtiya_list)
  - print(Adtiya)
  - print(Adtiya[1])
- Output:

```
>
==== RESTART: C:\Users\User\AppData\Local\Progra
('bob', 'Adtiya')
Adtiya
>
```

5. Aim: To Write Various Program For Manipulating Sets By Adding, Deleting, Elements Within The Set.

- Input:
  - Aakash= {'bob','Marry'}
  - Aakash.add('Sachin')
  - print(Aakash)
  - set1={'Isha','Aakash'}
  - set1.remove("Isha")
  - print(set1)
- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Pyth
{'Sachin', 'Marry', 'bob'}
{'Aakash'}
>>
```

6. Aim: To Run Various Commands For Manipulating The Data Of A Dictionary.

- Input:

a. Deleting in a dictionary

- people={ 'jack':30, 'mary':26}
- people.pop('jack')
- print(people)

- Output:

```
>>>
==== RESTART: C:\Users\User\AppData\Local\Programs\I
{'mary': 26}
>>>
```

- Input:

b. Retrieving in a dictionary

- people={ 'pop':30 , 'Mary':25}
- pop_age=people['pop']
- print(pop_age)

- Output:

```
>>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\
30
>>>
```

- Input:

c.Adding in a dictionary

- people={‘bob’:30,’Mary”:25}
- people[‘Sachin’]=32
- print(people)

- Output:

```
>>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Pyt
{'bob': 30, 'Mary': 25, 'Sachin': 32}
>>>
```

- Input:

d. Check if a given key already exist in the dictionary

- people={ 'pop':30 , 'Mary':25}
- if 'pop' in people:
- print (' pop exist in the dictionary ')
- else :
- print('pop does not exist in the dictionary ')

- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Python313\i
 pop exist in the dictionary
>>
```

7. Aim: To Write A Program To Understand Loops Over Loops By Printing Every Element In The List By Using Nested For Loops.

- Input:
  - Car_models=[['BMW I8', 'BMW X3 ', 'BMW X1 '],
  - ['Ferrari B12 ', ' FERRARI F8 ' , 'FERRARI GTC4 '],
  - ['MCLAREN 570S','MCLAREN570GT ',' MCLAREN 720S']]
  - for brand in Car_models:
  - for model in brand :
  - print(model)
- Output:

```
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Pythor
BMW I8
BMW X3
BMW X1
Ferrari B12
 FERRARI F8
FERRARI GTC4
MCLAREN 570S
MCLAREN570GT
 MCLAREN 720S
```

8. Aim: To Write A Program For Running Loops Over Other Data Structures.

- Input:

a. Loop over a tuple

- people=('jack' , 'Mary ')
- for person in people :
- print(person)

- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Pytho
jack
Mary
>>
```

- Input:

b. Loops over set

- people={'sachin' , 'Manisha '}
- for person in people :
- print(person)

- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Python3
Manisha
sachin
>>
```

- Input:

c. Loops over dictionary

- people={ 'jack':30 , 'Mary':25}
- for person in people:
- print(person)

- Output:

```
>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Pyth
jack
Mary
>
```

9. Aim: To Perform Comparison Of Two Numbers Using Ternary Operator.

- Input:
  - n1=int(input("Enter the first number :"))
  - n2=int(input("Enter the second number :"))
  - result=n1 if n1>n2 else n2
  - print(result)
- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\
Enter the first number :30
Enter the second number :35
35
>>
```

10. Aim: To Understands Various Types Of Scopes By Writing Their Respective Program.

- Input:

a. Global scope

- name="Aakash"
- def print_name():
- print("my name is " + name)
- print_name()

- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Pytho
my name is Aakash
>>
```

- Input:

b. Local scope

- name = "Sachin"
- def print_name():
- name = "Aakash"
- print("my name is " + name)
- print_name()

- Output:

```
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Python
my name is Sachin
>
```

11. Aim: To Write A Program Of A Calculator To Accept Whether The User Wants To Continue Or Not.

- Input:
  - def add(x=10,y=20):
  - return x+y
  - def sub(x=20,y=40):
  - return x-y
  - def multiplication(x=28,y=22):
  - return x*y
  - def divide (x=30,y=70):
  - return x/y
  - ch ="y"
  - def cal():
  - a = int(input("enter a number:"))
  - b = int(input("enter a number:"))
  - c = int(input("enter a choice 1 -add, 2-sub,3-multiplication,4-divide:"))
  - if c == 1:
  - print(add())
  - elif c == 2:
  - print(sub())
  - elif c == 3:
  - print(multiplication())
  - elif c == 4:
  - print(divide())
  - else:
  - print("no search found")
  - while ch =="y":
  - cal()
  - ch = input("want to continue y/n:")
- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Python313\i
no search found
enter a number:23
enter a number:24
enter a choice 1 -add, 2-sub,3-multiplication,4-divide:225
want to continue y/n:25
>>
```

12. Aim: To Write A Program In Order To Accept Multiple Parameters Under Single Argument.

- Input:
  - def Aakash(*rags):
  - for variable in rags:
  - print(variable)

  - Aakash(1,2,3,4,5,6,)
- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Pyth
1
2
3
4
5
6
>>
```

13. Aim: To Write A Program In Order To Accept Multiple Parameters Under Multiple Arguments.

- Input:
  - def Aakash(**kwargs):
  - for key, value in kwargs.items():
  - print(key)
  - print(value)
  - Aakash(first=10, second=20)
- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\P
first
10
second
20
>>
```

14. Aim: To Perform Various Task Using Lambda Function.

a. Lambda function with one parameter

- Input:
  - Aakash = lambda variable: variable**3
  - print(Aakash (3))
- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Python313\input.py =
27
>>
```

b. Lambda function with multiple parameters

- Input:
  - Aakash = lambda number1, number2, number3: number1*number2*number3
  - print(Aakash (2,3,4))
- Output:

```
>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Python313\input.py ===
24
>
```

c. Map function

- Input:
  - Aditya = [2,3,4]
  - a=list(map(lambda xyz: xyz**3,Aditya))
  - print(a)
- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Python313\input.py
[8, 27, 64]
>>
```

15. Aim: To Create An Array.

- Input:
  - import array as arr
  - arr=arr.array('i',[10,20,30])
  - print(len(arr))
- Output:

```
>>
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Python313\input.py ==
3
>>
```

16. Aim: To Perform Various Operations On 1D And 2D Arrays.

- Input:
  - import numpy as np
  - arr=np.arange(1,10,2)
  - print("elements of array: ", arr)
  - arr1=arr[np.array([4,0,2,-1,-2])]
  - print("indexed element of array arr: ",arr1)
  - arr2 =np.arange(4)
  - print("array arr2: ", arr2)
  - print("the element At the index 0 of arr2 is: ",arr2[0])
  - print("the element at the second index of aar2 is: ",arr2[2])
  - arr3=np.arange(12)
  - arr4=arr3.reshape(3,4)
  - print("array arr4:\n",arr4)
  - print("element at 0th row and 0th column of arr4 is:", arr4[00])
  - print("element at 1st row and 2nd column of arr4 is: ", arr4[1,2])
  - print("array arr4:\n",arr4)
  - print("\n")
  - print("1st row: \n: ",arr4[1])
  - print(arr3)
  - print("element at index 6 of an array arr: ", arr3[6])
  - print("element from index 3 to index 8 are of an array arr3: ", arr3[3:8])
- Output:

```
==== RESTART: C:\Users\User\AppData\Local\Programs\Python\Pythor
elements of array : [1 3 5 7 9]
indexed element of array arr : [9 1 5 9 7]
array arr2: [0 1 2 3]
the element At the index 0 of arr2 is : 0
the element at the second index of aar2 is : 2
array arr4:
 [[ 0 1 2 3]
 [ 4 5 6 7]
 [ 8 9 10 11]]
element at 0th row and 0th column of arr4 is : [0 1 2 3]
element at 1st row and 2nd column of arr4 is : 6
array arr4 :
[[ 0 1 2 3]
 [ 4 5 6 7]
 [8 9 10 11]]
1st row :
: [4 5 6 7]
[ 0 1 2 3 4 5 6 7 8 9 10 11]
element at index 6 of an array arr: 6
element from index 3 to index 8 are of an array arr3 : [3 4 5 6 7]
```